

Solving Rational Equations Worksheet

Quadratics:

6 Solve for x :

a $\frac{x+1}{4} = \frac{1}{2x}$

b $\frac{x+4}{2} = \frac{6}{x}$

c $\frac{x+2}{x} = x$

d $\frac{x-1}{x+2} = \frac{2}{x}$

e $\frac{x}{1+2x} = \frac{1}{3x}$

f $\frac{3x+1}{2x} = x+2$

7 Solve for x :

a $\frac{2}{x} + \frac{3}{x+2} = 1$

b $\frac{5}{x} + \frac{x}{x-6} = 2$

c $\frac{4}{x+1} + \frac{3}{x+3} = -1$

d $\frac{5}{x+2} - \frac{6}{x-1} = -2$

e $\frac{4}{x-2} + \frac{3}{x-1} = 3$

f $\frac{x}{x-2} - \frac{x+3}{x} = -5$

g $\frac{1}{x+3} + \frac{1}{x-3} = x$

h $\frac{4}{x+2} - \frac{2}{x+1} = x$

Practice 1

Solve these rational equations, leaving your answers exact, or to 3 s.f.

1 $\frac{x}{4} + \frac{3x}{5} = 2$

2 $\frac{5x-9}{6} + \frac{2x-4}{7} = 1$

3 $\frac{2x}{7} - \frac{1}{3} = \frac{x}{2}$

4 $\frac{x+3}{6} + \frac{2x}{5} = \frac{5x}{2}$

5 $\frac{2x-1}{4} - \frac{x+4}{3} = \frac{5x}{6}$

6 $\frac{4x+1}{2} + \frac{3x-5}{5} = \frac{2x-2}{4}$

7 $\frac{x-3}{2} + \frac{3x-1}{4} = \frac{4x+7}{6} + \frac{1-5x}{8}$

8 $\frac{5x}{3} - \frac{4x-1}{8} = \frac{x+2}{6} - \frac{x+1}{12}$

Problem solving

9 $\frac{2}{x} - \frac{3x}{4} = \frac{1}{3}$

Practice 3

Solve these rational equations using appropriate equivalence transformations, and check your answers algebraically. Make sure you check for extraneous solutions.

$$1 \quad \frac{4}{x} = \frac{9}{x-2}$$

$$2 \quad \frac{10}{x+4} = \frac{15}{4(x+1)}$$

$$3 \quad \frac{6}{3x} + \frac{5}{4} = \frac{3}{x}$$

$$4 \quad \frac{4}{3x} + \frac{5}{4} = \frac{3}{x}$$

$$5 \quad \frac{10}{x(x-2)} + \frac{4}{x} = \frac{5}{x-2}$$

$$6 \quad \frac{5}{x-2} = 7 + \frac{10}{x-2}$$

$$7 \quad \frac{3}{x-2} + \frac{2}{2-x} = 2 - x$$

$$8 \quad x + 1 = \frac{72}{x}$$

$$9 \quad x + \frac{3}{x+1} = 4$$

Solutions:

- 6** **a** $x = -2$ or 1 **b** $x = -6$ or 2 **c** $x = 2$ or -1
 d $x = 4$ or -1 **e** $x = 1$ or $-\frac{1}{3}$ **f** $x = \frac{1}{2}$ or -1
7 **a** $x = 4$ or -1 **b** $x = 15$ or 2 **c** $x = -9$ or -2
 d $x = 3$ or $-\frac{7}{2}$ **e** $x = 4$ or $\frac{4}{3}$ **f** $x = 1$ or $\frac{6}{5}$
 g $x = 0$ or $\pm\sqrt{11}$ **h** $x = 0$ or -3

Practice 1

- 1** $\frac{40}{17}$ **2** $\frac{129}{47}$
3 $-\frac{14}{9}$ **4** $\frac{15}{58}$
5 $-\frac{19}{8}$ **6** 0
7 $\frac{73}{29}$ **8** $\frac{3}{26}$
9 $1.43, -1.87$ (3 s.f.)

Practice 3

- 1** $-\frac{8}{5}$ **2** $\frac{4}{5}$ **3** $\frac{4}{5}$ **4** $\frac{4}{3}$
5 No solutions, $x = 2$ is an extraneous solution
6 $\frac{9}{7}$
7 No solutions,
8 $8, -9$
9 $3.303, -0.303$ (3 d.p.)